Algorithmic Illusion? An Empirical Study of Generative AI Image Ban on a Food Delivery Platform

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Abstract

Platforms increasingly use generative AI to create and enhance content, but it remains unclear whether such images boost user engagement or erode consumer trust. We investigate this question in the context of India's food delivery market. Following the surge of generative AI technologies, India's two largest food delivery platforms—Zomato and Swiggy—began offering generative AI tools to help restaurants beautify dish images starting in late 2023. However, in August 2024, Zomato enacted a platform-wide ban on AI-generated images, citing concerns over authenticity, while its main rival, Swiggy, continued to allow their use. Using a unique panel of consumers' purchases across both platforms from March 2024 to February 2025, we implement a difference-in-difference design with Zomato restaurants as the treated group and Swiggy restaurants as the control group. We find that, following the ban, Zomato restaurants experienced significantly greater increases in order volume, revenue, and customer counts, compared with restaurants on Swiggy. Consumers also became more willing to try previously unvisited restaurants on Zomato. Multi-modal image forensic analyses and three state-of-the-art generative AI detectors consistently show that Swiggy's dish images were more likely to be AI-generated after the policy divergence. Moreover, Swiggy restaurants with more "AI-like" images experienced larger sales declines compared with Zomato restaurants. Overall, our findings suggest that banning generative AI-modified dish images may restore perceived authenticity, rebuild consumer trust, and enhance platform competitiveness. These results highlight the need for platforms and merchants to exercise caution when deploying generative AI in contexts where authenticity is critical to consumer decision-making.

Keywords Generative AI, food delivery, platform competition, algorithm aversion, image forensics

1 Introduction

The rapid development of generative artificial intelligence (Generative AI) has created both opportunities and challenges for businesses and society (Demirci et al., 2025; Shan and Qiu, 2025). While the technology holds great potential to enhance productivity and user experiences, it also raises concerns regarding misinformation, authenticity, and consumer protection (Campante et al., 2025). Policymakers around the world have begun to intervene: for example, the Japanese parliament recently passed a law to curb the "rapid spread of false or harmful content generated by AI tools" and granted the government authority to publicly name companies found misusing such technologies.[1]

In parallel, the food delivery industry has grown rapidly over the past decade, reshaping how consumers interact with restaurants (Li and Wang, 2025). Platforms such as Uber Eats, Deliveroo, Foodpanda, Meituan, Zomato, and Swiggy act as intermediaries between restaurants and consumers, enabling convenient online transactions. Uber Eats alone increased its revenue from 1.4 billion USD in 2019 to 12.2 billion USD in 2023.[2] To sustain growth and attract customers in this highly competitive sector, platforms continually experiment with technological innovations to improve user experience.

A recent wave of innovation in the industry comes from the adoption of generative AI. Food delivery platforms are experimenting with AI-powered assistants to guide consumers during the purchase process[3] and with AI-modified dish images that make menus more visually appealing.[4] These applications promise to enhance browsing and shopping experiences. However, they also raise critical questions about consumer trust and perceptions of authenticity. If dish images diverge too far from the food actually delivered, customers may feel misled, avoid reordering from the same restaurant, or even reduce their engagement with the platform. Moreover, concerns about inauthentic images may discourage consumers from trying new restaurants altogether. On Reddit, there have been numerous posts complaining that the dish images shown online differ from the food customers actually receive. Many users also report avoiding restaurants whose dish images appear inauthentic or even boycotting Swiggy as a platform [5]. Thus, while generative AI offers exciting opportunities for platforms, the impact of having generative AI-manipulated content on consumer behavior and market outcomes remains uncertain. If consumers rely on the authenticity of certain content, such as dish images, when making purchase decisions, having generative-AI content may even backfire. Hence, our research question is, *What is the impact of (banning) generative-AI dish images on food delivery*

[1] See `https://babl.ai/japan-passes-new-ai-law-balancing-innovation-and-risk-mitigation/` (accessed August 2025).

[2] See `https://fourweekmba.com/uber-eats-revenue/` (accessed December 2024).

[3] See `https://tinyurl.com/4cezb3ne` (accessed December 2024).

[4] See `https://tinyurl.com/38zsv8tv` (accessed December 2024).

[5] See `https://www.reddit.com/r/mildlyinfuriating/comments/1euhwvl/ai_pics_on_swiggy_are_misleading/` and `https://www.reddit.com/r/swiggy/comments/1nkf3lv/swiggy_should_restrict_restaurants_putting_ai/` (accessed November 2025).

platform order volume and revenue? How do consumers react to restaurants having images that are more likely to be generative-AI-generated?

We conduct our empirical analysis using data from India's two largest food-delivery platforms, Zomato and Swiggy, which together generated approximately 5 billion USD in revenue in 2024.[6] In late 2023, following the launch of ChatGPT, both platforms introduced generative AI–based image enhancement tools that allowed restaurant owners to improve dish quality, adjust aesthetics, and choose from a variety of background options.[7] Restaurants could also employ external tools such as Mid-Journey to generate, modify, or blend images.[8] While Swiggy continued permitting restaurants to use AI-generated images throughout 2024 and 2025, Zomato imposed a platform-wide ban on August 18, 2024, citing consumer complaints that AI-modified images were misleading and risked eroding trust. Restaurants that violated the policy faced de-listing from the platform.[9] Figure 1 shows how one Swiggy restaurant utilizes generative AI to display and represent dish images.[10]

Figure 1: One Swiggy Restaurant Disclosing Gen AI Usage



This policy divergence between Zomato and Swiggy offers a unique quasi-experimental setting to study the impact of generative AI on consumer behavior and platform competition. On one hand, AI-enhanced images may attract consumers by improving visual appeal and lowering search costs. On the other hand, if consumers perceive these images as deceptive, they may reduce trust in restaurants or the platform itself. The contrasting strategies of the two platforms allow us to investigate whether the benefits of image enhancement outweigh the potential costs of eroding authenticity, and how such trade-offs shape consumer demand and competitive dynamics in online food delivery markets.

Utilizing a unique dataset documenting a random sample of more than 13,000 consumers' purchase records on the two major food-delivery platforms in India from March 2024 to February 2025, we construct a restaurant-level sales panel. Treating Zomato restaurants as the "treated" group and Swiggy restaurants as

[6] See `https://tinyurl.com/63m7s2ns` and `https://www.jika.io/quote/ZOMATO.NS/financials/revenue` (accessed July 2025).

[7] See `https://tinyurl.com/49mbspbu` and `https://tinyurl.com/2e2atrty` (accessed June 2025).

[8] See `https://tinyurl.com/2tbz5sne` (accessed December 2024).

[9] See `https://tinyurl.com/4yr24ffs` (accessed June 2025).

[10] See `https://www.swiggy.com/city/kolkata/mums-cafe-n-resto-belghoria-rest821126` (accessed October 2025).

the control group, and applying a difference-in-differences approach, we find that restaurants' order volume and revenue on Zomato increased significantly compared with those on Swiggy after the policy change. We also observe that an increasing number of consumers shifted to purchasing from Zomato relative to Swiggy and became more willing to try new restaurants on Zomato. This pattern suggests that eliminating concerns around generative-AI dish images may help restore consumers' confidence in exploring new establishments on the platform.

We further strengthen our analysis by conducting a series of image-forensic tests and multi-modal analyses using state-of-the-art generative-AI image detectors. Forensic examinations of image sharpness, compression-artifact error levels, and GAN fingerprint residuals provide consistent evidence that Swiggy's dish images are more likely to be AI-generated. Three HuggingFace-based generative-AI detectors yield similar results. We also find that restaurants with a higher likelihood of using AI-generated images experience larger declines in sales, reflecting consumers' dislike towards images that are more "AI-like" (Exner et al., 2025).

This study makes several contributions and has important policy implications. First, we contribute to the literature on human-AI interaction (Castelo et al., 2019; Yang et al., 2025; Krakowski et al., 2025). "algorithm aversion" theory Castelo et al. (2019) posits that humans would prefer humans over algorithms. We use real-world consumer purchase records and a natural experiment to reveal consumers' preference for real dish images over AI-generated dish images, which can inform platform decision-making and merchants' image selection.

Secondly, we contribute to the growing empirical studies on consumers' perception of generative AI content. Since the launch of ChatGPT in late 2022, generative AI has been transforming the business models of enterprises in many industries. Yet, research exploring consumers' reactions to the content, especially images, generated by AI in real-world practice has been limited (Exner et al., 2025; Li et al., 2025). Exner et al. (2025) implements a field experiment and shows that, AI-generated ads receive more clicks compared with human-generated images only if these AI-generated images do not "look like AI". Our empirical findings are consistent with their results: restaurants whose dish images have lower AI likelihood experience smaller sales declines, whereas those with higher AI likelihood suffer more substantial reductions in order volume.

2 Dataset and Model-Free Evidence

2.1 Dataset

Our research combines two datasets from different sources. The first is a unique dataset of 13,755 consumers' food delivery purchase records on Zomato and Swiggy in India, obtained from Measurable AI, a leading

market intelligence company specializing in tracking consumers' platform transactions. The dataset covers the period from March 1, 2024, to February 28, 2025, and includes purchase records from the same set of consumers across both platforms. It contains rich information such as order value, discounts (Swiggy only), restaurant name, dish purchased, dish price (Swiggy only), and restaurant address. A key feature of this dataset is its ability to observe individual consumers' behavior across platforms, enabling direct cross-platform comparisons. This feature is particularly valuable when implementing our cross-platform analyses.

The second dataset consists of restaurant dish images that we collected from Zomato and Swiggy after Zomato's generative AI ban in Kolkata, one of the largest cities in India. We merged the consumer purchase record dataset (restricted to Kolkata) with the restaurant image dataset using restaurant name and address when conducting the mechanism analysis and multi-modal image analysis. Because restaurants on Zomato are the entities directly affected by the policy change, we aggregate consumers' order volumes to obtain restaurant-level sales, which we use as a proxy for restaurant demand.

We present summary statistics of restaurants in Table 1. Overall, restaurants on the two platforms are comparable, though Zomato hosts slightly more restaurants, while Swiggy restaurants have marginally higher pre-policy order volumes and lower average order values.[11]

Table 1: **Sample Descriptive Statistics of Restaurants**

Platform	Variable	Count	Mean	Median	Min	Max
Swiggy	Chain (=1)	30,763.00	0.39	0.00	0.00	1.00
Swiggy	Pre-Policy Order Volume	30,763.00	1.19	1.00	0.00	92.00
Swiggy	Pre-Policy Order Value (INR)	18,343.00	322.83	253.00	0.00	6,080.00
Swiggy	Multihoming	30,763.00	0.12	0.00	0.00	1.00
Zomato	Chain (=1)	33,021.00	0.39	0.00	0.00	1.00
Zomato	Pre-Policy Order Volume	33,021.00	1.10	1.00	0.00	49.00
Zomato	Pre-Policy Order Value (INR)	19,068.00	342.86	270.02	19.36	11,039.09
Zomato	Multihoming	33,021.00	0.11	0.00	0.00	1.00

2.2 Model-Free Evidence

Figure 2 plots average order volume per restaurant on Zomato and Swiggy from March 2024 to February 2025 in India and in Kolkata, where we have restaurant dish image information. Before August 2024, which is the month when Zomato banned the use of generative-AI–created dish images (marked by the red line), the two platforms exhibited parallel trends in both average order volume and total orders.

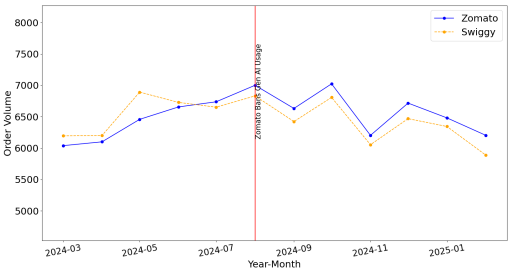
After the ban, however, Zomato's trend diverged upward. Restaurants on Zomato experienced a notice-

[11] Pre-Policy Average Order Value is missing for restaurants with zero pre-policy orders.

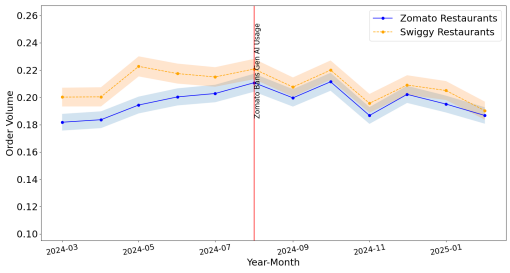
able and persistent increase in order activity, while Swiggy's performance plateaued or slightly declined. The magnitude of this divergence is visible both at the all-India level and within Kolkata.

Together, these visual patterns provide preliminary, model-free evidence consistent with a positive causal impact of Zomato's AI-image ban on consumer engagement and restaurant demand. The pre-treatment parallel trends further support the identification assumptions underlying our subsequent difference-in-differences analysis.

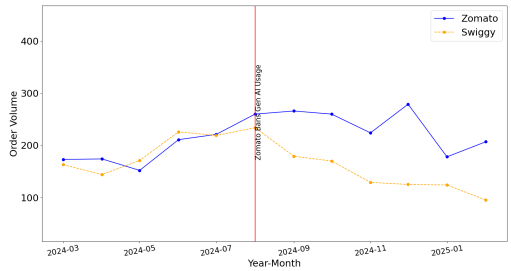
Figure 2: Model-Free Evidence on Platform Performance



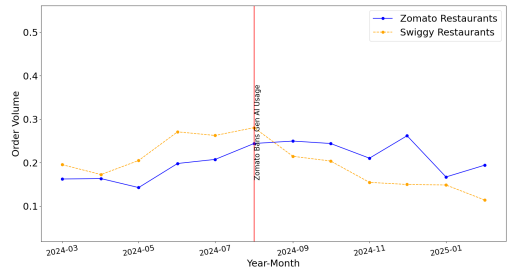
(a) Total number of orders on two platforms in India



(b) Average order volume per restaurant on two platforms in India



(c) Total number of orders on two platforms in Kolkata



(d) Average order volume per restaurant on two platforms in Kolkata

3 Regression Analysis

3.1 Restaurants' Order Volume and Revenue

We conduct a difference-in-difference analysis to assess the impact of the generative AI image ban on restaurants' order volume. The control group is restaurants on Swiggy, while the treatment group is restaurants on Zomato.

The regression model that we use is,

$$\mathrm{Y}_{rt} = \alpha + \beta_1 Treat_r \times PostPolicy_t + \delta_r R_r + \gamma_t T_t + \varepsilon_{rt} \tag{1}$$

where Y_{rt} is restaurants' weekly order volume, total revenue, and number of customers. $Treat_r$ indicates whether a restaurant is a Zomato restaurant or not, $PostPolicy_t$ indicates whether week t is after the policy change or not, R_r is restaurant fixed effect, and T_t is week fixed effect to control seasonality and time trend. Note that for multi-homing restaurants that join both platforms, we treat them as different restaurants on each platform. This is because the same restaurant may offer different dishes, the same dishes at different prices, and, importantly, use different images for the same dish—especially given that they may continue to use generative AI to modify their dish images on Swiggy. The results are displayed in Table 2.

Table 2: **Impact of the Policy Change on Restaurants' Order Volume and Revenue**

	(1) Order Volume	(2) Revenue
Treat × PostPolicy	0.00229***	0.966***
	(0.000756)	(0.286)
Constant	0.0454***	15.40***
	(0.000214)	(0.0811)
Rest FE	Yes	Yes
YearWeek FE	Yes	Yes
Observations	3380552	3380552
R^2	0.085	0.078

Notes: Robust standard errors, clustered at the restaurant level, are presented in parentheses.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

We find that after Zomato banned restaurants from using generative AI, restaurants on the platform experienced a 5.1% increase in order volume and 6.2% increase in revenue compared to Swiggy. This suggests that consumers likely value authenticity and consistency between dish images and the delivered food more than the visual appeal of the images themselves. These findings indicate that the generative AI ban may enhance Zomato's competitiveness.

3.2 Restaurants' Order Source Analysis

The increase in Zomato restaurants' order volume could stem from two sources. The first is new customers to these restaurants. Previously, with the use of generative AI to modify dish images, customers may have been reluctant to try unfamiliar restaurants because of the risk that the food would look different from the online images. Following the platform-level ban on generative AI for dish images, Zomato consumers may have regained trust in the authenticity of images, making them more willing to explore new restaurants.

The second source is returning customers. In the past, a customer might have been drawn to a restaurant with appealing images but, after receiving food that did not match the photos, chosen not to reorder. With the generative AI ban on dish images, there should be improved consistency between images and actual

dishes on Zomato. If customers enjoy the food, they are more likely to reorder from the same restaurant. Nevertheless, this remains an empirical question worth further investigation.

To examine this, we conduct a subgroup analysis categorizing restaurant orders by source—new versus returning customers.[12] To measure traffic to restaurants, we sum the orders from new and returning customers for restaurant r in week t, defining them as $OrderVolumeNewCustomers_{rt}$ and $OrderVolumeReturningCustomers_{rt}$, respectively. These two measures serve as dependent variables in Equation 1. The regression results are presented in Table 3.

Table 3: **Order Source Analysis**

	(1) Order Volume New Customers	(2) Order Volume Returning Customers
Treat × PostPolicy	0.000885**	0.00141**
	(0.000369)	(0.000610)
Constant	0.0258***	0.0196***
	(0.000105)	(0.000173)
Rest FE	Yes	Yes
YearWeek FE	Yes	Yes
Observations	3380552	3380552
R^2	0.014	0.132

Notes: Robust standard errors, clustered at the restaurant level, are presented in parentheses.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

We find that the policy change has a significant positive impact on Zomato restaurants in attracting both new customers' and returning customers' orders. This suggests that Zomato's generative AI ban boosts consumer traffic by encouraging them to explore new restaurants and also fueling returning purchases.

3.3 Restaurants' Customer Number

We further analyze policy change's impact on consumer traffic. The dependent variables that we employ include number of total consumers, number of new consumers, and number of returning consumers. New and returning consumers are defined as discussed in the previous subsection. We re-run regression model 1 and results are presented in Table 4. We find that the policy change has a positive impact on number of total, new, and returning consumers, in line with the order source analysis results.

[12] We define a new customer as a first-time buyer to restaurant t; otherwise, they are classified as a returning customer to restaurant t.

Table 4: **Consumer Source Analysis**

	(1) Num Total Customers	(2) Num New Customers	(3) Num Returning Customers
Treat × PostPolicy	0.00202***	0.000885**	0.00114***
	(0.000597)	(0.000369)	(0.000423)
Constant	0.0416***	0.0258***	0.0159***
	(0.000169)	(0.000105)	(0.000120)
Rest FE	Yes	Yes	Yes
YearWeek FE	Yes	Yes	Yes
Observations	3380552	3380552	3380552
R^2	0.069	0.014	0.120

Notes: Robust standard errors, clustered at the restaurant level, are presented in parentheses.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

4 Multi-modal Analysis

4.1 Forensic Tests

To assess whether restaurant dish images were likely generated or enhanced by generative AI, we also conducted a series of forensic tests on a large sample of images collected from Swiggy and Zomato.

We applied three complementary forensic techniques: (i) Error Level Analysis (ELA), which detects inconsistent compression artifacts; (ii) Noise and Frequency Spectrum analysis, which evaluates sharpness and high-frequency texture patterns; and (iii) GAN fingerprint residuals, which identify statistical anomalies characteristic of synthetic image generation. These methods are widely used in digital image forensics and provide triangulated evidence of AI-based modification (Dzanic et al., 2020; Neves et al., 2022). Figure 3 illustrates these tests for a representative cake image from the same restaurant on Swiggy and Zomato[13]. The Swiggy version shows inconsistent error levels across the cake surface and background props, extremely high sharpness (≈ 225) with dense high-frequency textures, and residual statistics with heavy-tailed distributions (mean = 1.70) and higher ELA (95th percentile is 50). In contrast, the Zomato version exhibits lower sharpness (≈ 113), softer frequency patterns consistent with casual smartphone photography, residuals aligned with typical image compression (mean = 1.29), and lower ELA (95th percentile is 40). Taken together, these differences strongly suggest that the Swiggy image was AI-generated or AI-enhanced (e.g., background replacement, synthetic texture sharpening), while the Zomato image is more likely to be authentic.

Similarly, another group pair is Chicken Biryani provided by a restaurant named Akram's [14], as shown in

[13]The restaurant name is Cakes, and the dish name is Dry Choco Fudge Cake. Their URLs on the two platforms are `https://www.zomato.com/kolkata/cakes-1-beliaghata/order` and `https://www.swiggy.com/city/kolkata/cakes-beliaghata-rest58090` (accessed September 2025).

[14]Their URLs on the two platforms are `https://www.zomato.com/kolkata/akrams-kestopur/order` and `https://www.swiggy.com/city/kolkata/akrams-kestopur-lake-town-rest779306` (accessed September 2025).

Figure 4. The swiggy image shows much higher sharpness (1,808 vs. 147), indicating denser high-frequency detail and potential oversharpening from generative or editing tools. They also exhibit higher mean pixel intensity (6.80 vs. 2.60) and greater variation (standard deviation of 9.20 vs. 2.28), suggesting higher dynamic range manipulation and contrast enhancement. The ELA metrics are likewise higher for Swiggy (mean = 41.6, 95th percentile = 150) compared to Zomato (mean = 30.1, 95th percentile = 90), consistent with repeated recompression or partial image synthesis.

Extending this analysis to the full sample of restaurant dish images (Swiggy: 35,083 unique images; Zomato: 51,044 unique images), we find systematic differences across all three forensic measures. On average, Swiggy dish images exhibit substantially higher sharpness (921 versus 622), higher residual statistics (mean of 3.87 versus 3.40), and higher error level analysis (95th percentile of 42.64 versus 34.87). These patterns suggest that Swiggy images are consistently more processed or enhanced relative to their Zomato counterparts.

These results are consistent with our prediction that Swiggy restaurants adopted generative AI-modified images, while Zomato banned them. The discrepancy across platforms is large and robust enough to serve as a useful falsification test in our empirical analysis.

Figure 3: Images of "Dry Choco Fudge Cake" from *Cakes* (Left Swiggy, Right Zomato)

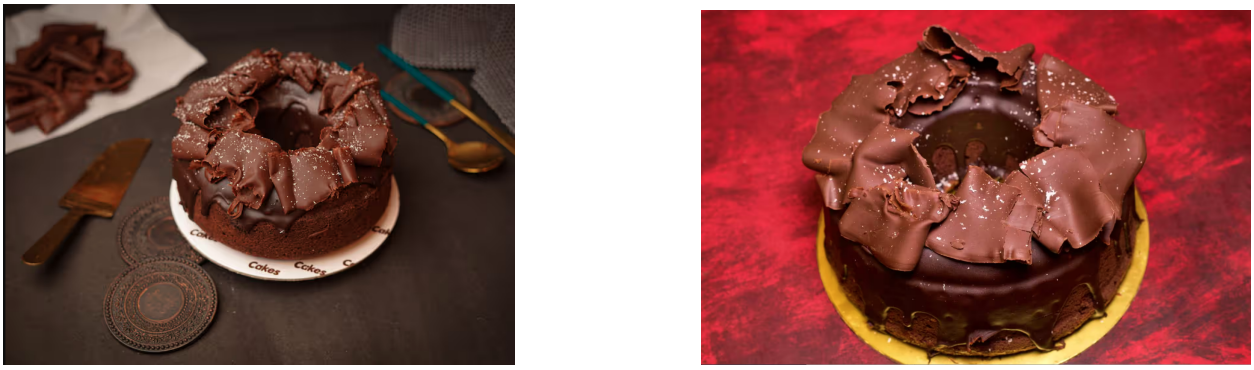


Figure 4: Images of "Chicken Biryani" from *Akram's* (Left Swiggy, Right Zomato)



4.2 Generative AI Image Detection Analysis

To assess whether dish images on Swiggy are more likely to be generated or modified by generative AI relative to Zomato, we applied three independent HuggingFace detection models ("umm-maybe/AI-image-detector," "Smogy/SMOGY-Ai-images-detector," and "Ateeqq/ai-vs-human-image-detector") to all dish images from restaurants on the two platforms. Each detector was applied in batch mode. For each image, we extracted the probability score of being AI-generated ("AI prediction score"). We then averaged these scores across all available dish images within a restaurant, yielding one aggregated score per restaurant per platform. Note that, while the generative AI image detector tends to be reliable, they are not completely accurate (Exner et al., 2025). Our expectation is that, the detectors would predict dish images on Swiggy to be significantly more likely to be AI-generated or AI-modified compared with Zomato, while acknowledging the possible existence of type-I and type-II errors for individual dish images. For example, "umm-maybe/AI-image-detector" claims that its prediction accuracy is around 0.94 on its validation set [15].

We first compared these restaurant-level AI prediction scores between Swiggy and Zomato, separately for two groups of multi-homing restaurants: those displaying *different* images across the two platforms and those displaying *identical* images. For restaurants with different images, at the dish image level, of all the 3153 Swiggy dish images, $median_{swiggy} = 0.33$, and of all the 2342 Zomato dish images, $median_{zomato} = 0.04$. At the restaurant level, the average AI prediction score is significantly higher for Swiggy compared to Zomato ($N = 34$, $t = 5.64$, $p < 0.001$, $median_{swiggy} = 0.32$, $median_{zomato} = 0.15$). This result is consistent with our prediction that Swiggy is more likely to adopt AI-modified or AI-generated dish images. In contrast, for restaurants that display identical images across the two platforms, the difference is statistically insignificant ($N = 144$, $t = -0.95$, $p = 0.34$). For single-homing restaurants, we find similar results ($N_{Swiggy} = 478$,$N_{Zomato} = 797$, $t = 10.25$, $p < 0.001$).[16]

Taken together, these results provide evidence that the observed discrepancies in dish images between Swiggy and Zomato are systematically related to the use of generative AI, rather than random variation in photography or presentation style. In particular, when a restaurant reuses the same image across platforms, there is no systematic difference in AI likelihood, but when images differ, Swiggy's version is much more likely to carry generative AI signatures.

[15] See `https://huggingface.co/umm-maybe/AI-image-detector`.

[16] Restaurants with missing AI scores on Swiggy are excluded from the analysis. Average AI score is missing for restaurants without dish images and restaurants whose URLs are not available. Among all the 1277 Zomato restaurants, 247 of their URLs are no longer available. Among all the 1251 Swiggy restaurants, 494 of their URLs are no longer available. The average AI score for each restaurant is computed from three independent HuggingFace detectors.

5 Mechanism Analysis

5.1 Heterogeneous Effects by Dish Image AI Likelihood

To further investigate whether the post-policy effects differ depending on the extent of AI involvement in restaurant imagery, we use the median Swiggy restaurant-level AI prediction score (0.34) as a cutoff to classify restaurants into *low-* and *high-AI-likelihood* groups. All Zomato restaurants are retained in both regressions as the control group, while Swiggy restaurants are included in the low- or high-AI-likelihood sample depending on their average AI score. This treatment group design is a flip side of Equation 1 as we intend to capture the differential effect across Swiggy restaurants, compared with Zomato restaurants, which are not allowed to use Generative AI to modify dish images.

Table 5 reports the results. For restaurants with low AI-likelihood images (Column 1), the coefficient on the interaction term *Swiggy* × *PostPolicy* is smaller than less significant ($\beta = -0.0185$, $p < 0.05$), suggesting that the policy change affect them less. In contrast, for restaurants with high AI-likelihood images (Column 2), the coefficient is more negative and more statistically significant ($\beta = -0.0221$, $p < 0.01$), indicating a more noticeable decline in order volume following the generative-AI policy.

Overall, these results strengthen our interpretation that the negative post-policy impact on Swiggy is more concentrated among restaurants whose dish images are more likely to have been generated or altered by AI, with a less salient effect on restaurants appear "less AI-like". Consumers appear to respond more unfavorably to such AI-enhanced imagery once the use of generative AI becomes a salient concern.

Table 5: **Moderating Effect of Dish Image AI Likelihood**

	Low-AI-Likelihood Restaurants	High-AI-Likelihood Restaurants
	(1)	(2)
	Order Volume	Order Volume
Swiggy × PostPolicy	-0.0185**	-0.0221***
	(0.00909)	(0.00779)
Constant	0.0490***	0.0493***
	(0.000808)	(0.000698)
Restaurant FE	Yes	Yes
YearWeek FE	Yes	Yes
Observations	66886	66992
R^2	0.064	0.067

Notes: Robust standard errors, clustered at the restaurant level, are presented in parentheses.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

5.2 Multi-homing and Single-homing Restaurants

Theoretically, multi-homing restaurants, which are restaurants available on both platforms, have less incentive to use generative AI images on Swiggy for consistency reasons. In addition, consumers may also verify dish images across platforms for multihoming restaurants to check the authenticity of dish images. Therefore, even if they intended to beautify dish images on Swiggy, the effect would be limited or null. As a result, policy change should not have a significant impact on their order volume. By contrast, single-homing restaurants, which are restaurants that are available on only one platform, are more likely to employ generative AI images on Swiggy as they do not have consistency concerns, and consumers are not able to verify dish images across platforms. These restaurants are likely to be more affected by the policy change. We re-run regression model 1 based on restaurants' multi-homing status, and the results are displayed in Table 6, which are in line with our predictions.

Table 6: **Order Volume Analysis by Restaurants' Multihoming Status**

	Single-Homing Restaurants	Multi-Homing Restaurants
	(1)	(2)
	Order Volume	Order Volume
Treat × PostPolicy	0.00247***	0.000922
	(0.000781)	(0.00267)
Constant	0.0431***	0.0633***
	(0.000222)	(0.000747)
Rest FE	Yes	Yes
YearWeek FE	Yes	Yes
Observations	2996090	384462
R^2	0.081	0.104

Notes: Robust standard errors, clustered at the restaurant level, are presented in parentheses.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

5.3 Alternative Control Group: Multi-homing Restaurants with Different Dish Images

One concern is that factors other than dish images may drive the observed changes in order volume across platforms. While we cannot directly observe whether a restaurant used Generative AI to modify its dish images, we address this issue by exploiting variation among multi-homing restaurants. Specifically, we construct an alternative control group of Swiggy restaurants that display different dish images from those shown on Zomato.

Theoretically, when a restaurant is listed on both platforms, it has incentives to use the same images for

consistency and to avoid additional photography costs. Therefore, if a restaurant displays different images on Swiggy compared to Zomato after the policy change, it is plausible that the Swiggy images were modified using Generative AI.

In total, we identify 190 multi-homing restaurants. Among them, 37 (19.62%) display different dish images across the two platforms. We use these 37 restaurants as the control group and re-estimate Equation 1 for the three dependent variables mentioned, namely, order volume, total revenue, and number of customers. The regression results are presented in Table 7.

Table 7: **Alternative Control Group - Swiggy Restaurants that Have Different Dish Images**

	(1) Order Volume	(2) Revenue	(3) Num Customers
Treat × PostPolicy	0.0470**	18.57***	0.0331**
	(0.0188)	(6.503)	(0.0161)
Constant	0.0215**	9.919***	0.0254***
	(0.0101)	(3.486)	(0.00861)
Restaurant FE	Yes	Yes	Yes
YearWeek FE	Yes	Yes	Yes
Observations	57187	57187	57187
R^2	0.063	0.089	0.054

Notes: Robust standard errors, clustered at the restaurant level, are presented in parentheses.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

5.4 Alternative Control Group: Multi-homing Restaurants that have Same Dish Images

Theoretically, if restaurants use the same dish images across platforms, their order volumes should not be affected by the policy change. We therefore use the 149 Swiggy multi-homing restaurants that display identical dish images on both Swiggy and Zomato as the control group and re-estimate Equation 1. The regression results are presented in Table 8.

Table 8: **Alternative Control Group - Swiggy Restaurants that Have the Same Dish Images**

	(1) Order Volume	(2) Revenue	(3) Num Customers
Treat × PostPolicy	0.0000540	0.136	-0.00180
	(0.0102)	(5.863)	(0.00889)
Constant	0.0476***	20.47***	0.0451***
	(0.00515)	(2.972)	(0.00451)
Restaurant FE	Yes	Yes	Yes
YearWeek FE	Yes	Yes	Yes
Observations	60473	60473	60473
R^2	0.065	0.091	0.057

Notes: Robust standard errors, clustered at the restaurant level, are presented in parentheses.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

5.5 Correlation between Swiggy Restaurant Ratings and AI Likelihood

Anecdotal observations suggest that consumers tend to respond negatively to restaurants whose dish images appear to be AI-generated [17]. To provide quantitative support for this pattern, we conduct a correlation analysis between the AI likelihood score of restaurant images on Swiggy and their corresponding average customer ratings. After removing missing values, we find a statistically significant negative association. The Pearson correlation between the AI likelihood score and Swiggy ratings is $-0.068(N = 661, p = 0.079)$, and the Spearman rank correlation is $-0.091(N = 661, p = 0.019)$[18]. These results indicate that restaurants with higher probabilities of using AI-generated images tend to receive slightly lower customer ratings, consistent with the notion that consumers may perceive such images as less authentic and unfavorable.

6 Robustness Checks and Alternative Explanations

6.1 Parallel Trend Assumption Check

The difference-in-difference analyses rely on the parallel trend assumption. To check the validity of the assumption, following Aral and Dhillon (2021), we create a series of week dummies to indicate the week 1, 2, 3, ..., 12+, before the policy change. We then interact each of the dummy variables with the treatment variable and run the following regression:

$$Y_{rt} = \alpha + \sum_{\tau=-7,\ldots,-1} \beta_\tau 1[t - event_t = \tau] \times Treat_r + \delta_r R_r + \gamma_t T_t + \varepsilon_{jt} \quad (2)$$

[17] See https://www.reddit.com/r/swiggy/comments/1nkf3lv/swiggy_should_restrict_restaurants_putting_ai/ (accessed November 2025).

[18] Spearman rank correlation is more robust to outliers (Sedgwick, 2014).

where Y_{rt} includes restaurants' weekly order volume and total revenue. $event_t$ represents the week in which the policy shock occurred. $1[t - event_t = \tau]$ are a series of "event-time" dummies that equal one when the policy shock is τ periods away from event t. The week before the policy shock serves as the baseline, and its event-time dummy variable is omitted. The regression results are shown in 9. As shown in the regression results, the coefficients are not significantly different from 0, indicating that Swiggy and Zomato restaurants do not have a significant difference between order volume and revenue, lending support for our difference-in-difference analysis.

Table 9: **Robustness Check Showing Parallel Trends in the Difference-in-Difference Specifications**

	(1) Order Volume	(2) Total Revenue
Week 1 × Treat	-0.000883	0.0795
	(0.00202)	(0.844)
Week 2 × Treat	-0.00143	0.523
	(0.00269)	(1.116)
Week 3 × Treat	-0.00000889	0.500
	(0.00268)	(1.114)
Week 4 × Treat	0.000226	-0.206
	(0.00264)	(1.119)
Week 5 × Treat	-0.00207	-0.147
	(0.00262)	(1.105)
Week 6 × Treat	0.000935	0.321
	(0.00259)	(1.107)
Week 7 × Treat	0.000532	-0.276
	(0.00258)	(1.112)
Week 8 × Treat	-0.00192	0.143
	(0.00261)	(1.122)
Week 9 × Treat	0.000487	0.428
	(0.00256)	(1.095)
Week 10 × Treat	0.00446*	1.042
	(0.00253)	(1.087)
Week 11 × Treat	0.00180	0.124
	(0.00252)	(1.105)
Constant	0.0457***	14.99***
	(0.000961)	(0.408)
Rest FE	Yes	Yes
YearWeek FE	Yes	Yes
Observations	1530816	1530816
R^2	0.131	0.113

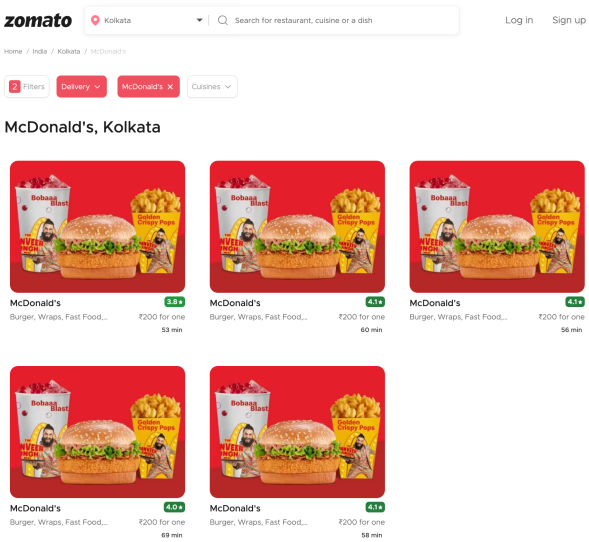
Notes: Robust standard errors, clustered at the restaurant level, are presented in parentheses.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

We find that before the policy change, there is no significant difference between Zomato and Swiggy restaurants' order volume, lending support to the parallel trend assumption check.

6.2 Chain Restaurants

Theoretically, while small and independent restaurants have more freedom and flexibility to utilize generative AI technology, chain restaurants have less flexibility when it comes to altering or improving their images. This is because dish images for chain restaurants are typically designed and standardized by headquarters. Moreover, these chain restaurants may be more cautious about employing generative AI to enhance their visuals. For instance, all McDonald's outlets on Zomato in Kolkata display the same header and dish images, as shown in Figure 5. Similarly, the fast-food behemoth uses identical header and dish images across its branches on Swiggy. This consistency suggests that McDonald's likely does not rely on generative AI for its food photography.

Figure 5: Sample Pages of McDonald's in Kolkata on Zomato



In addition to the low likelihood of chain restaurants using generative AI, customers are usually more familiar with the food offered by chain restaurants. Even if these restaurants do use generative AI, the impact on customer purchases is likely limited, as customers already have clear expectations about the food they will receive when ordering online. Therefore, we predict that the policy change would not significantly affect chain restaurants' order volumes, but would primarily impact non-chain restaurants. To test this, we run a separate regression using Equation 1, splitting the sample into chain and non-chain restaurant subgroups. The results are displayed in Table 10.

We find that the regression results align with our prediction. The policy change has no significant impact on chain restaurants, while the number of orders from non-chain restaurants has increased significantly. These

Table 10: **Analysis on Restaurants' Chain Status**

	Chain Restaurants	Non-Chain Restaurants
	(1) Order Volume	(2) Order Volume
Treat × PostPolicy	0.00119	0.00298***
	(0.00128)	(0.000930)
Constant	0.0532***	0.0405***
	(0.000364)	(0.000263)
Rest FE	Yes	Yes
YearWeek FE	Yes	Yes
Observations	1304065	2076487
R^2	0.094	0.077

Notes: Robust standard errors, clustered at the restaurant level, are presented in parentheses.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

results have two implications. First, they empirically show that chain restaurants are less likely to adopt and benefit from the use of generative AI technology for image modification. Second, banning the use of generative AI technology improved non-chain restaurants' order volume and market share. This result may seem counterintuitive, as small restaurants often lack the resources and access to professional photographers to take high-quality dish photos, and generative AI technology could ostensibly help them. However, it appears that consumers value authenticity more, and banning the use of generative AI ultimately across the platform benefits non-chain restaurants and improves their competitiveness.

The insignificant results for chain restaurants also rule out the possibility that platform-level promotional activities or unobserved time trends are driving the changes in food delivery order volume and regression results. Otherwise, chain restaurants should have a significant increase after August 2024 as well.

6.3 Order Value

We perform an additional robustness analysis to rule out the possibility that the observed increase in order volume is driven by temporary promotions or platform- or restaurant-level pricing changes. Specifically, we re-estimate Equation 1 using (i) the restaurant's average order value and (ii) the average number of dishes per order as dependent variables. If the post-policy increase in order volume were primarily due to promotions (e.g., "1-for-1" deals or discounts), we would expect to observe lower average order value or more dishes per order.

As shown in Table 11, the estimated treatment effects on both average order value and number of dishes per order are statistically insignificant. These results suggest that the policy change did not affect order size or value, reinforcing that the observed rise in order volume on Zomato is unlikely to be driven by promotional

activities or price adjustments.

Table 11: **Impact of the Policy Change on Rests' Order Value**

	(1) Avg Order Value
Treat × PostPolicy	-4.129
	(3.970)
Constant	348.4***
	(1.120)
Restaurant FE	Yes
YearWeek FE	Yes
Observations	101260
R^2	0.574

Notes: Robust standard errors, clustered at the restaurant level, are presented in parentheses.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

6.4 Promotions and Dish Price on Swiggy

To ensure that the observed changes in platform choice are not confounded by concurrent price or promotion adjustments, we further conduct a robustness check examining whether Swiggy's average order discounts and average dish prices exhibit any discontinuous changes around the policy implementation week.[19] Specifically, we estimate the Regression Discontinuity Design (RDD) model:

$$Y_{rt} = \beta_0 + \beta_1 \text{Weeks to Policy Change}_t + \beta_2 \text{PostPolicy}_t + \beta_3 (\text{Weeks to Policy Change}_t \times \text{PostPolicy}_t) + \alpha_r + \varepsilon_{rt}, \tag{3}$$

where Y_{rt} denotes either the average order discount amount (in Indian Rubee) or average dish price for restaurant r in week t, Weeks to Policy Chang$_t$ is the running variable measuring weeks relative to the policy cutoff, and α_r captures restaurant fixed effects. The coefficient β_2 represents the discontinuity at the cutoff, indicating whether Swiggy changed its discount or pricing policy immediately after the policy took effect.

Table 12 reports the results. The estimated discontinuities at the cutoff are small and statistically insignificant for both outcomes: for the average discount, $p = 0.408$, while for the average dish price, $p = 0.164$. Neither effect is statistically significant at conventional levels. These results suggest that Swiggy did not systematically adjust its discounting or pricing strategies around the policy implementation period.

Taken together, this robustness check supports our identification strategy by confirming that the observed shifts in consumer platform choices are unlikely to be driven by concurrent changes in Swiggy's pricing or

[19] Given we do not have promotion information on Zomato, we cannot run a DiD analysis on these two variables.

discount policies, but rather by the Zomato generative AI ban policy intervention itself.

Table 12: **Robustness Check on Avg Order Discounts and Avg Dish Price**

	(1) Avg Discount Amount Swiggy	(2) Avg Dish Price Swiggy
Weeks to Policy Change	1.280	-4.073
	(2.286)	(3.195)
PostPolicy=1	-0.409	9.365
	(5.060)	(7.071)
PostPolicy=1 × Weeks to Policy Change	-3.261	4.501
	(2.582)	(3.621)
Constant	114.5***	197.8***
	(4.252)	(6.271)
Restaurant FE	Yes	Yes
Observations	2491	3728
R^2	0.750	0.743

Notes: Robust standard errors, clustered at the restaurant level, are presented in parentheses.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

6.5 User-Level Regression

So far, we have been presenting restaurant-level regression results to demonstrate the policy change's impact on restaurant sales. To check if a user-level regression would yield similar results, we implement a simple reduced-form model:

$$Y_{it} = PostPolicy_t + U_i + \varepsilon_{it} \tag{4}$$

where Y_{it} includes user i's monthly number of order difference between Zomato and Swiggy (number of Zomato orders minus number of Swiggy orders), which measures the order volume change between platforms, U_i is user fixed effect, and T_t is year-month fixed effect. The results are displayed in Table 13. Consistent with previous results, at the platform level, we find that food delivery users shifted their orders from Swiggy to Zomato.

Table 13: **Impact of the Policy Change on Users' Consumption on Two Platforms**

	Order Volume Zomato minus Order Volume Swiggy	
	(1)	(2)
	All Users	Users whose Pre-Policy Order Volume is Larger than 0
PostPolicy	0.0254*	0.0648***
	(0.0145)	(0.0197)
Constant	-0.0105	-0.0148
	(0.00844)	(0.0115)
User FE	Yes	Yes
Observations	163896	115848
R^2	0.608	0.623

Notes. Robust standard errors are clustered at the user level and reported in parentheses.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

7 Conclusions and Contributions

This research has several contributions. The first one is that the paper adds to existing literature discussing the trade-off between algorithm usage and consumer trust. There has been extensive literature discussing the usage of algorithms and consumers' perceptions, for example, Kim and Duhachek (2020). We contribute to the literature by analyzing the new technology advancement of generative AI on food delivery consumer perceptions.

Secondly, we also explored the potential unequal benefits of technology advancement on business entities, for example, by analyzing the potential unequal order volume change between large chain restaurants and small independent restaurants. Consumers may also react differently to business entities' technology adoption based on their expectations.

Thirdly, we also contribute to the literature on platform competition from the perspective of technology usage (Armstrong, 2006). We explored how the ban on generative AI affects platform competition and marketplace dynamics.

Fourthly, our research contributes to business studies in emerging markets, particularly South Asia. As one of the largest markets in the world, India and similar economies remain underexplored in terms of empirical research on business models and consumer behavior due to data limitations. Such research is critically important given the region's growing influence in the global market.

Our research is not without limitations. One key limitation is that we do not observe which restaurants used generative AI to modify dish images prior to the policy change. While this limitation may not affect the validity of our cross-platform analysis, future research with more detailed datasets could explore potential spillover effects in order volume changes between restaurants that adopted the technology and those that

did not after a generative AI ban.

Appendix

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